Experiment 6: Dimensionality Reduction and Model Evaluation (With and Without PCA)

Objective

To study the effect of **dimensionality reduction** using Principal Component Analysis (PCA) on the performance of various machine learning classifiers. The task requires:

a) Training and validating models without PCA (original feature space).

b) Training and validating models with PCA (reduced feature space).

For both cases, students must perform hyperparameter tuning, apply 5-fold cross-validation, and record performance.

Dataset

Students should clearly state:

- Dataset source (e.g., UCI, Kaggle).
- Number of samples and features.
- Target classes and distribution.
- Preprocessing steps (standardization, encoding, handling missing values).

Models to Evaluate

1. Support Vector Machine (SVM)
2. Naïve Bayes
3. k-Nearest Neighbors (KNN)
4. Logistic Regression
5. Decision Tree
6. Random Forest
7. AdaBoost
8. Gradient Boosting
9. XGBoost
10. Stacking (base learners + meta-learner)

Procedure

1. Preprocess and standardize features.
2. Apply PCA (students choose variance target or fixed number of components).
3. For each model, define a hyperparameter search grid (students decide ranges).
4. Train and validate each model under both No-PCA and With-PCA settings.
5. Use 5-fold cross-validation and record results.
6. Compare and analyze whether PCA improved accuracy, F1-score, or stability.

PCA Summary

Table 1: PCA Variance Explained

Setting	Chosen Components / Variance Target	Explained Variance (%)	Justification
With-PCA			

Hyperparameter Tuning Templates (Examples)

Support Vector Machine (SVM)

Table 2: SVM — Hyperparameter Tuning Results

Kernel	C Values Tried	Gamma Values Tried	Performance (No-PCA)	Performance (With-PCA)

Naïve Bayes

Table 3: Naïve Bayes — Smoothing Choices

Smoothing Parameter	Performance (No-PCA)	Performance (With-PCA)

Table 4: KNN — Hyperparameter Tuning

k **Values**	**Weights**	**Distance Metrics**	**Performance (No-PCA)**	**Performance (With-PCA)**

KNN

Students should create similar tables for: Logistic Regression, Decision Tree, Random Forest, AdaBoost, Gradient Boosting, XGBoost, and Stacking.

Cross-Validation Results (All Models)

Table 5: 5-Fold Cross-Validation Results (No-PCA vs With-PCA)

Model	**Fold 1**	**Fold 2**	**Fold 3**	**Fold 4**	**Fold 5**	**Avg (No-PCA)**	**Avg (With-PCA)**
SVM Naïve Bayes KNN Logistic Regression Decision Tree Random Forest AdaBoost Gradient Boosting XGBoost Stacking							

Observation Questions

- Which models improved most with PCA? Which did not? Why?
- Did PCA reduce variance across folds (more stable results)?
- For high-dimensional data, was PCA beneficial in reducing overfitting?
- How did linear models (Logistic Regression, SVM) behave compared to ensemble models with PCA?
- Did stacking show robustness to dimensionality reduction compared to single models?

Report Checklist

- Dataset details and preprocessing.
- PCA design choice (components or variance target) with scree plot.
- Hyperparameter grids defined for each model.
- Completed fold-wise tables and average metrics (No-PCA and With-PCA).
- ROC/PR curves and confusion matrices for selected models.
- A final conclusion on when PCA helps and when it does not.